

Module 10: Technology Transfer Models and Strategies

Technology Transfer Models

Technology Transfer (TT) means sharing technology from one place (like a university, research centre, or company) to another (like an industry or startup). The goal is to bring new ideas to the market, encourage innovation, and help society. Different models are used for this.

1. Linear Model

- **Description:** Technology moves step by step from research to the market.
- **Phases:** Research → Development → Production → Marketing.
- **Example:**
A university creates a new medicine → tests it in clinical trials → gives a license to a pharma company → the company produces and sells it.

2. Market-Oriented Model

- **Description:** Research is started only after knowing what customers or the market needs.
- **Key Idea:** Market demand drives research.
- **Example:**
A smartphone company surveys users → finds interest in foldable screens → develops flexible OLED → launches foldable phones.

3. Open Innovation Model

- **Description:** Many groups (universities, companies, startups, even customers) work together and share ideas/technology.
- **Example:**
Procter & Gamble's "Connect + Develop" program, where outside researchers and inventors help create new products.

4. Triple Helix Model

- **Description:** Universities, industries, and governments work together for innovation.
- **Example:**
Government funds a university's renewable energy project → university develops the technology → a company produces and sells it with government support.

5. Network Model

- **Description:** Several organizations (research labs, companies, incubators) form a network to develop and transfer technology.

- **Example:**
European Union Horizon projects where different universities and companies work together on new technology.

6. Push-Pull Model

- **Description:**
 - **Technology-Push:** Research creates new tech and pushes it into the market.
 - **Market-Pull:** Market demand encourages new technology development.
- **Example:**
 - **Technology-Push:** A lab develops 5G tech and encourages telecom companies to use it.
 - **Market-Pull:** Growing demand for electric cars leads to better battery research.

7. Licensing Model

- **Description:** The technology owner allows others to use it under certain conditions (like paying royalties or fees).
- **Example:**
Microsoft licenses Windows OS to computer manufacturers.

8. Joint Venture Model

- **Description:** Two or more companies work together to create and sell new technology.
- **Example:**
Sony and Ericsson joined to form Sony Ericsson, sharing mobile technology knowledge.

University–Industry Collaboration (UIC):

University–Industry Collaboration means partnerships between universities, colleges, and research institutes with industries like companies, startups, and factories. The purpose is to use each other's strengths and connect theory with real-world practice.

Objectives of University–Industry Collaboration

1. **Knowledge Transfer** – Sharing university research results with industry for practical use.
2. **Skill Development** – Helping students gain job-ready skills through industry exposure.

3. **Innovation & R&D** – Doing joint research to create new products, processes, or technologies.
4. **Resource Sharing** – Using labs, equipment, and expertise from both sides.
5. **Economic Development** – Supporting local and national growth through innovation.

Modes of Collaboration

- **Internships & Training** – Students gain work experience in companies.
- **Joint Research Projects** – Universities and companies develop technology together.
- **Sponsored Research** – Industry pays universities to research in a certain field.
- **Consultancy Services** – Professors give expert advice to industries.
- **Incubation Centres & Startups** – Universities provide space and support for industry-backed startups.
- **Guest Lectures & Workshops** – Industry experts teach or train students.

Benefits

For Universities:

- Extra research funding.
- Real-world topics for academic work.
- More job opportunities for students.

For Industry:

- Access to university talent.
- Cost-effective innovation.
- Collaboration on advanced technology.

Example of University–Industry Collaboration

Case Study: IIT Bombay & Tata Consultancy Services (TCS)

- **Goal:** Research in Artificial Intelligence (AI) and Machine Learning (ML).
- **Actions:**
 - TCS funded an AI research centres at IIT Bombay.
 - Jointly developed AI algorithms for natural language processing.
 - Students got internships and job offers from TCS.
- **Result:**
 - Published quality research papers.
 - Developed commercial AI solutions.
 - Improved student skills and employability.

Government and Public-Private Partnerships

Governments and PPPs (Public-Private Partnerships) play a major role in technology transfer. They create policies, funding, and platforms to help move technology from research labs to industries and end users.

1. Government Role in Technology Transfer

Governments often act as **supporters, regulators, and funders**. They make policies, fund research, and give incentives for commercialization.

Government Initiatives

- **Policies & Regulations:** Setting up patent laws, IPR systems, and innovation policies.
- **Funding:** Providing grants, subsidies, and tax benefits for R&D.
- **Infrastructure:** Building technology parks, incubators, and labs.
- **Licensing:** Government labs license their technologies to private companies.

Example 1: National Innovation Foundation (India)

- **Goal:** Support local/grassroots innovations and help transfer them to industries.
- **How it works:** Finds local inventions, patents them, and partners with companies.
- **Case:** A farmer's low-cost seed sowing machine patented and licensed to an agri-equipment company.

Example 2: NASA Technology Transfer Program (USA)

- NASA gives its patents to private companies for use in health, cars, and energy.
- **Case:** NASA's memory foam for aircraft seats is now used in mattresses and pillows.

2. Public-Private Partnerships (PPP) in Technology Transfer

PPP means joint collaboration between government and private companies to develop and commercialize technology.

Key Features of PPP:

- Shared investment in research.
- Shared risks and profits.
- Faster movement of technology to market.
- Better use of public research.

Types of PPP Models

- **Build-Operate-Transfer (BOT):** Private company builds technology, runs it for a while, then hands it to government.
- **Joint Ventures:** Government and private firms share ownership.
- **Contractual Partnerships:** Specific agreements for developing or commercializing technology.

Example 1: BIRAC (India)

- Government program for biotech innovation through PPP.

- Supported startups in vaccine development with research institutions.

Example 2: Fraunhofer Institutes (Germany)

- Funded by both government and private companies.
- Famous for creating the MP3 audio compression technology.

Benefits of Government and PPP in Technology Transfer

- Faster innovation.
- Less financial risk for private companies.
- Public access to important technologies (health, defense, energy).
- Job creation and economic growth.

Startups, Spin-offs, and Incubators

1. Startups in Technology Transfer

Definition: A startup is a new company that creates innovative products or services, often based on research or new technology. In technology transfer, startups bring university or lab research to the market.

Key Features:

- Based on innovative technology.
- High growth and high risk.
- Funded by venture capital, angel investors, or government grants.
- Common in AI, biotech, fintech, and clean energy.

Examples:

- **DeepMind (UK):** AI research startup later acquired by Google.
- **CureVac (Germany):** Biotech startup using mRNA tech, later important in COVID-19 vaccines.

2. Spin-offs in Technology Transfer

Definition: A spin-off is a new company created from a parent organization (university, research lab, or company) to commercialize specific technology.

Key Features:

- Comes from parent organization's research.
- Usually involves patented or unique knowledge.
- Often keeps a link with the parent (licensing, equity).
- Very common in universities.

Examples:

- **Google:** Started as a research project at Stanford University, licensed PageRank algorithm.
- **Arm Holdings (UK):** Spin-off from Acorn Computers to develop RISC processors.

3. Incubators in Technology Transfer

Definition: An incubator supports new startups or spin-offs by giving them space, mentorship, business services, and sometimes funding. In tech transfer, they help move research into market-ready products.

Key Features:

- Provides infrastructure and guidance.
- Helps in prototypes, IP protection, and business development.
- Often linked with universities or government.
- Reduces risk and improves chances of success.

Examples:

- **Y Combinator (USA):** Supported companies like Dropbox and Airbnb.
- **Stanford Research Park (USA):** Supports startups from Stanford research.

How They Interconnect in Technology Transfer

- **Startups** turn new technologies into products.
- **Spin-offs** come from universities/companies to take research to market.
- **Incubators** provide the support system for both to succeed.

Licensing vs. Direct Commercialization

1. Licensing

Licensing means the owner of a technology (called the **licensor**) gives another party (called the **licensee**) permission to use it. In return, the licensee pays fees or royalties. The owner still keeps the rights to the technology.

Key Features

- **Ownership:** The original creator keeps ownership of the technology.
- **Rights Granted:** The licensee is allowed to use, make, or sell the technology.
- **Revenue Model:** The licensor earns through royalties, one-time payments, or payments based on milestones.
- **Risk:** Lower risk for the licensor because the licensee handles production and sales.
- **Control:** The licensor has limited control, only as mentioned in the contract.

Types of Licensing

- **Exclusive License:** Only one licensee can use the technology.
- **Non-exclusive License:** Many licensees can use the same technology.
- **Cross-licensing:** Two parties give each other licenses for their technologies.

Example

- A university creates a patented drug formula.
- Instead of making the medicine itself, it licenses the patent to a company like Pfizer.
- Pfizer produces and sells the drug.
- The university earns royalties from sales but does not handle manufacturing or marketing.

2. Direct Commercialization

Direct commercialization means the creator of the technology makes, markets, and sells the product on their own, without giving rights to another company.

Key Features

- **Ownership:** The developer keeps full ownership and responsibility.
- **Revenue Model:** All income comes from selling products or services.
- **Risk:** High risk since the developer invests in production, marketing, and distribution.
- **Control:** Full control over product quality, branding, and pricing.
- **Capital Requirement:** Needs more money than licensing because of production and marketing costs.

Example

- A startup builds a new AI-powered translation device.
- Instead of licensing the technology to Sony, the startup makes and sells the device itself (online and in stores).
- The startup pays for production, marketing, and distribution, but keeps all the profit.